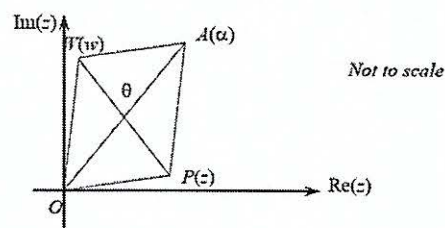
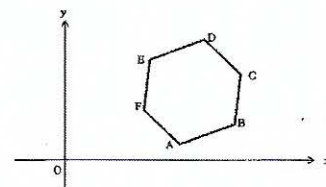


Complex Number 13

- Prove that $(1 + \cos \alpha + i \sin \alpha)^k + (1 + \cos \alpha - i \sin \alpha)^k = 2^{k+1} \cos\left(\frac{k\alpha}{2}\right) \cos^k\left(\frac{\alpha}{2}\right)$.
- Resolve $z^8 + 1$ into real quadratic factors.
 - Hence show that $\cos 4\theta = 8 \left(\cos \theta - \cos \frac{\pi}{8} \right) \left(\cos \theta + \cos \frac{\pi}{8} \right) \left(\cos \theta - \cos \frac{3\pi}{8} \right) \left(\cos \theta + \cos \frac{3\pi}{8} \right)$. (You do NOT need to prove $z^n + \frac{1}{z^n} = 2 \cos n\theta$ if $|z| = 1$ this time.)
- By expanding $(\cos \theta + i \sin \theta)^5$ show that $\tan 5\theta = \frac{5 \tan \theta - 10 \tan^3 \theta + \tan^5 \theta}{1 - 10 \tan^2 \theta + 5 \tan^4 \theta}$
 - Hence show that $\tan \frac{\pi}{5}$ is a solution of $z^4 - 10z + 5 = 0$
 - Hence find the exact value of $\tan \frac{\pi}{5}$.
- Show that $\frac{1 + \cos \theta + i \sin \theta}{1 - \cos \theta - i \sin \theta} = \frac{i \sin \theta}{1 - \cos \theta}$
 - Hence or otherwise show that $\frac{1 + \cos \theta + i \sin \theta}{1 - \cos \theta - i \sin \theta} = i \cot\left(\frac{\theta}{2}\right)$
 - Hence or otherwise show that if $z = i \cot\left(\frac{\theta}{2}\right)$ then $\frac{z-1}{z+1} = \cos \theta + i \sin \theta$
 - Hence solve the equation $\left(\frac{z-1}{z+1}\right)^4 = -1$.
- On the following diagram, $ABCDEF$ is a regular hexagon. If $\overrightarrow{OA} = z_1$ and $\overrightarrow{OB} = z_2$.
 - Express $\overrightarrow{AB}$, $\overrightarrow{BC}$ and $\overrightarrow{AD}$ in terms of z_1 and z_2
 - Show that $\overrightarrow{AB} \times \overrightarrow{BC} \times \overrightarrow{CD} = (z_1 - z_2)^3$
- Let $z = x + iy$ be any non-zero complex number.
 - Express $z + \frac{1}{z}$ in the form $p + iq$.
 - Given that $z + \frac{1}{z}$ is real, show that $y = 0$ or $x^2 + y^2 = 1$.
 - If $y = 0$, show that $\left|z + \frac{1}{z}\right| \geq 2$.
- Express $1 + i$ in mod-arg form.
 - Find the smallest positive value of m if $(1 + i)^m = (1 - i)^m$
- Find the nine roots of $z^9 - 1 = 0$, leaving your answers in mod-arg form.
 - If $P(z) = z^6 + z^3 + 1$, explain why the roots of $P(z) = 0$ are also roots of $z^9 - 1 = 0$.
 - Express $P(z)$ in form $P(z) = (z^2 - az + 1)(z^2 - bz + 1)(z^2 - cz + 1)$ where a , b and c are real numbers.
- The diagram shows a parallelogram $OPAW$ in the Argand plane. The diagonals intersect with an angle of θ as shown. Let z , α and w be the complex numbers represented by the points P , A and W respectively, where z , α and $w \neq 0$.
 - Given that $2\operatorname{Re}(\varphi) = \varphi + \bar{\varphi}$, where φ is a complex number, hence, or otherwise show that $2\operatorname{Re}(w\bar{z}) = w\bar{z} + \bar{w}z$.
 - Suppose $w\bar{z} + \bar{w}z = 0$, deduce that $\operatorname{Re}\left(\frac{w}{z}\right) = 0$.
 - Hence show that $OPAW$ is a rectangle.
 - Consider now that $\frac{w}{z} = ki$, where k is a real number,
 - Express $\frac{w-z}{w+z}$ in the form $a + ib$, where a and b are real numbers.
 - Hence find the expression for $\tan \theta$, where θ is the acute angle between the diagonals of $OPAW$ (assuming $k \neq 1$).



10. Use de Moivre's theorem to prove that: $(-\sqrt{3} + i)^n - (-\sqrt{3} - i)^n = 2^{n+1} \sin \frac{5\pi n}{6} i$
11. a. Use the identity $\cos 3\theta + i \sin 3\theta = (\cos \theta + i \sin \theta)^3$ to show that:
- $\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$
 - $\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$
- b. Show that $\tan 3\theta = \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta}$
12. Use similar methods to the previous question to show that:
- $\cos 4\theta = 8 \cos^4 \theta - 8 \cos^2 \theta + 1$
 - $\sin 4\theta = 4 \sin \theta \cos \theta (\cos^2 \theta - \sin^2 \theta)$
 - $\tan 4\theta = \frac{4 \tan \theta - 4 \tan^3 \theta}{1 - 6 \tan^2 \theta + \tan^4 \theta}$
13. If $z = \cos \theta + i \sin \theta$
- Show that $z - \frac{1}{z} = 2i \sin \theta$
 - Show that $z^n + \frac{1}{z^n} = 2 \cos n\theta$
 - Hence express $\sin^6 \theta$ in terms of $\cos 6\theta$, $\cos 4\theta$ and $\cos 2\theta$.
14. If $w = x + iy$ and $w = \frac{z-2i}{1-z}$
- Show that $z = \frac{x+i(y+2)}{(x+1)+iy}$
 - Find the locus of w if $|z| = 2$.
15. If w is the non real root of $z^7 - 1 = 0$ with the smallest possible positive argument:
- Show that $w = \cos \frac{2\pi}{7} + i \sin \frac{2\pi}{7}$ and that w^2, w^3, w^4, w^5 and w^6 are the other non real roots of $z^7 - 1 = 0$.
 - Show that $w + w^2 + w^3 + w^4 + w^5 + w^6 = -1$
 - By expanding $(w + w^6)(w^2 + w^5)(w^3 + w^4)$ show that $(w + w^6)(w^2 + w^5)(w^3 + w^4) = 1$.
Hence show that $\cos \frac{\pi}{7} \cos \frac{2\pi}{7} \cos \frac{3\pi}{7} = \frac{1}{8}$.
16. a. Simplify $(-4i)^3$.
- b. Hence write in modulus-argument form all the complex numbers z such that $z^3 = 64i$.
- c. If α and β are the two non-real cube roots of $64i$ with positive arguments prove that $\alpha^{12n} + \beta^{12n} = 2^{24n+1}$ where n is an integer.
17. The points A and B represent the complex numbers $z = 2 + i$ and $w = 2iz$ respectively. O is the origin.
- Show that $\angle AOB$ is a right angle.
 - If $OACB$ is a rectangle, find the complex number represented by C .

Answer

- $(z^2 - 2z \cos \frac{\pi}{8} + 1)(z^2 - 2z \cos \frac{3\pi}{8} + 1)(z^2 + 2z \cos \frac{3\pi}{8} + 1)(z^2 + 2z \cos \frac{\pi}{8} + 1)$
 - $c. \sqrt{5 - 2\sqrt{5}}$
 - $d. z = \pm i \cot \frac{\pi}{8}, \pm i \cot \frac{3\pi}{8}$
 - $a. \overrightarrow{AB} = z_2 - z_1, \overrightarrow{BC} = \text{cis} \frac{\pi}{3}(z_2 - z_1), \overrightarrow{AD} = 2 \text{cis} \frac{\pi}{3}(z_2 - z_1)$
 - $a. z + \frac{1}{z} = \frac{x(x^2+y^2+1)}{x^2+y^2} + \frac{y(x^2+y^2-1)}{x^2+y^2}i$
 - $1 + i = \sqrt{2} \text{cis}(\frac{\pi}{4})$
 - $m = 4$
- $z = \text{cis} 0, \text{cis} \frac{2\pi}{9}, \text{cis} \frac{4\pi}{9}, \text{cis} \frac{6\pi}{9}, \text{cis} \frac{8\pi}{9}, \text{cis} \frac{10\pi}{9}, \text{cis} \frac{12\pi}{9}, \text{cis} \frac{14\pi}{9}, \text{cis} \frac{16\pi}{9}$
 - $z^6 + z^3 + 1$ is a factor of $z^9 - 1$
 - $P(z) = (z^2 - 2z \cos \frac{2\pi}{9} + 1)(z^2 - 2z \cos \frac{4\pi}{9} + 1)(z^2 - 2z \cos \frac{8\pi}{9} + 1)$
 - $d. \frac{k^2-1}{k^2+1} + \frac{2k}{k^2+1}i, \tan \theta = \frac{2k}{k^2-1}$
 - $c. \sin^6 \theta = -\frac{\cos 6\theta}{32} + \frac{3 \cos 4\theta}{16} - \frac{15 \cos 2\theta}{32} + \frac{5}{16}$
 - $(x + \frac{4}{3})^2 + (y - \frac{2}{3})^2 = \frac{20}{9}$
 - $64i$
 - $z_1 = 4 \text{cis} \frac{\pi}{6}, z_2 = \text{cis} \frac{5\pi}{6}, z_3 = -4i$
 - $b. \overrightarrow{OC} = 5i$